



A Plagiarism Detector in Learning Management System for UTHM

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CHAPTER THREE

METHODOLOGY

This chapter explain the use of prototype model in this project and the activities that had been carried out in each phase.

3.1 Agile Model

The Agile Model is an iterative development process that creates prototype system versions that can be further enhanced through multiple iterations. Continuous delivery of user feedback helps to implement ongoing improvements to each new version of the prototype. The Agile approach emphasizes user satisfaction, collaborative processes, and flexibility in producing a final product that continues to improve over time by using the multiple refinements from iterations throughout the entire life cycle of the product. The Agile Model contains the following phases, Analysis, Design, Development, Testing, Deployment, Feedback, and the Iteration. Each phase is developed from the last, and iterations are utilized between phases to allow for continuous sharing between phases until the prototypes develop into the final system.

3.1.1 Planning Phase

During the planning phase of the project, the information was gathered on the project goals, objectives, and requirements through discussions with stakeholders including, lecturers, students, and UTHM IT portal team. The planning phase defines the requirements for the plagiarism detection module. For example, the ability to upload documents, check for similarities, manage users, and create reports. The project scope, project constraints, timeline and technologies required for the project are also established. The planning phase creates user stories or feature lists to represent what the system will be like from a user's perspective. User stories and lists will form the basis of future development cycles. The planning phase will help ensure that the development process is understandable on what will need to be developed prior to moving on.

3.1.2 Analysis Phase

To successfully design and develop a plagiarism detection in a Learning Management System, it is important to first conduct an analysis phase of each functional and technical requirement. The Analysis phase is necessary to determine the functional and technical requirements for a plagiarism detection module, and to prepare the system to meet the specific needs of its users, including students and lecturers. The analysis phase identifies student needs, including the ability for students to upload assignments with ease and receive notifications when their assignment has been submitted. Lecturer needs include automated similarity verification, detailed similarity reports, and ease of access to student documents. The analysis phase also determines the types of algorithms that will be used to process, compare, and validate submitted documents and defines how submitted documents will be processed, stored, and compared.

Analysis of current challenges experienced by AUTHOR, including the lack of an integrated plagiarism detection system, will be performed to ensure that the new system addresses these challenges and includes all the pertinent requirements. This analysis phase will also examine all of the technical limitations associated with implementing a plagiarism detection module, including accuracy of algorithms, server loads, and response time requirements, to ensure that the completed system meets all of these technical requirements. Upon conclusion of this analysis phase, a detailed system requirements list and workflow behaviour for the proposed plagiarism detection system will be published to assist with the subsequent design and development of the plagiarism detection module.

3.1.3 Design Phase

When creating the system design, all aspects of the system were organized so that it could function properly and be easy for both instructors and students to use. To do this, it was necessary to define the main components of the system, how the plagiarism detection module interacts with the submission process, and how the layout of the user interface will make it feel easy to navigate. It was also necessary to identify how the system handles certain functions, including file uploads, generating similarity reports, and providing alerts or notifications to users. Using Entity-Relationship Diagrams (ERD) and Data Flow Diagram (DFD), the database structure is designed

continuously. The design creates a comprehensive and organized plan to reference and ensure the system meets user needs.

3.1.4 Implementation Phase

The development of the system occurs in phases using an Agile Approach. A series of sprints will break down each function into smaller chunks. The main functions of the system that will be developed incrementally are User Login, Document Upload, Similarity Comparison, and Report Creation. Each chunk of development will be built, tested, and refined during a sprint until the final function is ready for production. In addition to these features, the algorithms used for detecting plagiarism will be created and integrated into the submission workflow for automatic analysis of any document uploaded.

Programming for backend functionality is done within the PHP and Python environment and database functionality is completed using MySQL. The frontend functionality is developed with HTML, CSS, and JavaScript. As each module is completed, stakeholders will have the opportunity to view how it works and provide input. If there are any recommended changes, they will be made before moving forward with the next sprint.

3.1.5 Testing Phase

The testing process is designed to verify that the System performs in accordance with the requirements specified and that it works correctly and efficiently. The first stage of testing is Unit Testing. During this stage each Module (e.g., File Uploading, Similarity Check Calculation, Report Generation) is tested separately to verify it is functioning as expected. The next stage of testing is System Integration Testing. During System Integration Testing, modules are tested together as a complete system and the flow of data between Modules is verified as being error-free. The final stage of testing is User Acceptance Testing (UAT). During UAT, Lecturers and Students test the system to assure themselves that it is easy to use, performs as intended, and satisfies the needs of its users. Each of the above tests contributes to creating a reliable, usable, and Deployable System.

3.2 System Development Workflow

There are a total of seven phases from the prototype model. As shown in Table 3.1, each phase has its assignment and output that needs to be produced during the entire project development. Besides that, the output had been completed within the specific days that were given.

Table 3.1: Software development activities and their task

Phase	Tasks	Outputs
Planning	• Define project objectives and scope • Identify users (lecturer & student) • Determine features and constraints • Develop project timeline and resource plan	• Project scope definition • Preliminary timeline
Analysis	• Gather system requirements from lecturer and students • Identify functional and non-functional requirements • Prepare use-case descriptions and workflow scenarios	• Software Requirements Specification (SRS) • List of system requirements • Develop Gantt chart
Design	• Develop system architecture and workflow • Design user interface layout and mockups	• UI mockups/layouts • System architecture design • Database schema and DFD

	• Plan database schema and data relationships	• Entity Relationship Diagram
Implementation	• Develop UI using chosen technology • Code system modules (submission, evaluation, feedback, etc.) • Integrate front-end and database • Conduct internal debugging	• Working prototype • Functional module integration • Source code files
Testing	• Conduct functional testing of each feature • Perform user acceptance testing with lecturer and students • Identify and log bugs/issues	• Test case results • UAT feedback forms • List of bugs and improvement items
Feedback & Iteration	• Collect detailed feedback from lecturers and students • Review issues, usability concerns, missing requirements • Refine and update prototype based on findings • Retest after modifications	• Improvement summary • Updated prototype version • Validation report after iteration

3.3 Chapter Summary

The methodology used to design the prototype plagiarism detection system for UTHM is detailed in this chapter. This chapter introduces the iterative approach selected because it enables ongoing improvements based on input from both instructors and students. Each phase of development is outlined, beginning with the planning phase where objectives and scope are established, then moving on to analysing the functional requirements and user requirements. The design phase provides an overview of the system architecture, interface design, and database schema that were all converted into a live prototype during implementation. Once the prototype was developed, testing was performed to ensure functionality and usability of the system and any problems found were addressed through the structured test and review cycle outlined in the methodology. This methodology allowed for the creation of a practical and user-friendly system that effectively meets the goals of Academic Integrity.